Formula 1
Podium Prediction Companion

ACL Milestone 1 - Analytical Report

Scope and main findings

This project combines 14 relational F1 tables to investigate race patterns and predict whether a driver will finish on the podium. The prediction cutoff is after qualifying and before the race starts. One modeling row represents one driver entering one race.

Cleaning and descriptive questions use all applicable historical records. Modeling uses 2003-2024 because qualifying availability is consistently high in this period. Earlier data supports historical form where available.

Final decision	Verified result
Selected model	Shallow neural network, hidden layers 32 and 16
Final inputs	9 numeric features and 8 missing-value flags
Preprocessing	Training-median filling, then training-fitted scaling
Decision threshold	0.43, selected using validation F1
Validation PR-AUC	0.7638
Unseen-test PR-AUC / F1	0.7078 / 0.6730
Reproducibility	193 code cells passed in a fresh local kernel

Reading guide

The next pages summarize cleaning, the three analytical answers, feature availability, models, experiments, XAI and inference. The figure appendix reproduces every static plot from the executed notebook in notebook order. Figure references identify the supporting evidence.

Interpretation boundary

Observed differences and XAI contributions describe associations or model behavior. They do not establish causal effects. Historical constructor conflicts, flagged timing/lap values, estimated starts and weak local LIME fidelity remain disclosed limitations.

1. Data Cleaning and Auditing

Raw CSV files remain immutable. Each dataset is inspected and cleaned on a copy, with before/after tables. The final audit checks non-null unique keys, foreign-key validity, remaining sentinel strings, semantic types and modeling grain. All implemented structural checks passed.

Dataset	Rows retained	Columns: before to after
results	26759	18 to 22
drivers	861	9 to 6
constructors	212	5 to 4
circuits	77	9 to 8
races	1125	18 to 7
status	139	2 to 2
seasons	75	2 to 1
qualifying	10494	9 to 12
driver_standings	34863	7 to 7
constructor_standings	13391	7 to 7
constructor_results	12625	5 to 4
sprint_results	360	16 to 19
lap_times	589081	6 to 7
pit_stops	11371	7 to 8

Main transformations

Loaded literal \N as missing; parsed dates and clock times; derived numeric duration columns. Results retain all 26,759 entries. Two supported missing positions were recovered, and 258 zero fastest-lap ranks became missing. Unavailable grids are missing in the analysis column. Sparse unused driver identifiers, session schedules and reference URLs were removed only from cleaned copies. Missing race outcomes/times were not invented.

Duplicate rule and open decisions

The outcome summary has 26,668 unique driver-race pairs. It independently takes the best recorded finishing order and best positive grid, which can come from different historical entries. Sixteen constructor-conflict groups, 12 flagged lap rows and 20 flagged timing rows remain unchanged pending TA guidance. Constructor conflicts lie outside the modeling period; disputed lap/timing fields are not final inputs. Figures 1-9 show initial and post-cleaning EDA.

2. Data-Engineering Questions

Question 1 - Which circuits convert front-row starts into podiums?

Joined unique outcomes to races and circuits. A front-row start is grid 1 or 2, rather than qualifying rank, because penalties can change the starting order. Missing grids are excluded. Conversion is podium finishes among front-row starters divided by front-row starts; each driver appearance counts once.

Across all years, Miami, Algarve, Las Vegas Strip, Donington Park, Ain Diab, Pescara and Mugello recorded 100% conversion, each based on only two to six starts. These perfect rates have little supporting evidence. Before 2014, Circuit of the Americas, Donington Park, Ain Diab and Pescara shared first place. From 2014, Miami, Algarve, Las Vegas Strip and Mugello shared first place; Suzuka followed at 88.89% (16/18).

Circuit	Before 2014	2014 onward	Rank: before to after
Suzuka	70.00%	88.89%	11 to 5
Yas Marina	60.00%	86.36%	25 to 8
Monaco	49.59%	85.00%	55 to 10
Circuit of the Americas	100.00%	80.00%	1 to 14

Rankings differ between periods, but the circuits and sample sizes also differ. Removing repeated historical entries changes some earlier estimates: Bremgarten rises from 63.64% to 87.50%. Modern estimates are unchanged. This is not evidence that the hybrid era itself caused the differences. See Figures 10-11.

Question 2 - Is there a home-country advantage at the same grid?

Joined outcomes, driver nationalities, races and standardized circuit countries. Dual nationalities count either country as home without duplicating appearances. East German maps to Germany; Rhodesian maps to Zimbabwe. We compare home/away rates within starting positions, then apply the same pooled grid weights to both groups.

Adjusted comparison	Podium rate
Home	13.24%
Away	13.60%
Home minus away	-0.36 percentage points

The adjusted comparison uses 25,021 appearances and does not show an overall home advantage. Rates vary by grid position. We did not test significance or control driver ability, team strength or season, so this is descriptive rather than causal. Figure 12 shows grid positions 1-10; the adjusted calculation uses all positions represented in both groups.

2. Data-Engineering Questions - Continued

Question 3 - Which constructors have the most mechanical retirements?

Joined results, race years, constructor names and status descriptions for 2014-2024. Entries marked withdrawn with zero completed laps are excluded, leaving 4,598 estimated starts. A documented list identifies component-related mechanical statuses; accidents, driver-related causes and regulation decisions are not mechanical failures. Unclear causes remain separate. Constructor IDs are kept distinct.

Mechanical-retirement rate = identified mechanical retirements / estimated starts x 100. Two cars from the same constructor contribute two starts. Figure 13 compares all constructor-period rates.

Period	Constructor	Failures / starts	Rate
2014-2021	Lotus F1	13 / 75	17.33%
2014-2021	Caterham	5 / 33	15.15%
2014-2021	Toro Rosso	35 / 242	14.46%
2014-2021	Renault	24 / 199	12.06%
2014-2021	McLaren	38 / 316	12.03%
2022-2024	Alfa Romeo	10 / 88	11.36%
2022-2024	Alpine F1 Team	14 / 135	10.37%
2022-2024	AlphaTauri	5 / 88	5.68%
2022-2024	Williams	7 / 134	5.22%
2022-2024	Ferrari	6 / 134	4.48%

Lotus F1 has the highest identified rate in 2014-2021; Alfa Romeo has the highest in 2022-2024. Rates are status-based estimates: the dataset cannot always distinguish a failure before the start from a first-lap retirement. There are 98 unclear causes in the earlier period and 22 in the later period. They remain in the denominator but not the identified-failure count; resolving them could change rankings.

Analysis grain and cutoff

All three questions are retrospective descriptive analyses, so their podium/failure outcomes are used to answer the questions. Modeling instead excludes current-race outcomes from its inputs. Missing qualifying sessions and incomplete race timings were retained rather than filled with invented observations.

3. Feature Selection, Engineering and Preprocessing

Coverage evidence supports modeling 2003-2024. Training-only plots and correlations justify grid, qualifying, recent form and pre-race standings. Better starting/qualifying positions, stronger recent podium performance and better championship positions are associated with more podium finishes in training. Figures 14-18 show this evidence.

Final numeric feature	Meaning	Availability
grid_for_analysis	Recorded positive starting position	Known at the pre-race cutoff
qualifying_position	Final qualifying position	Qualifying has finished
recent_podium_rate	Podium percentage over up to 5 earlier appearances	Current outcome shifted out
recent_avg_finish	Mean finishing order over the same earlier appearances	Current outcome shifted out
past_races_used	Number of earlier appearances used, capped at 5	History available before this race
driver_points_before_race	Driver championship points	After the previous race this season
driver_standing_before_race	Driver championship position	After the previous race this season
constructor_points_before_race	Constructor championship points	After the previous race this season
constructor_standing_before_race	Constructor championship position	After the previous race this season

Missing inputs are filled with training medians. Eight missing-value flags preserve whether a value was initially unavailable; past_races_used had no missing training values and has no flag. Nine numeric inputs are standardized using training means and standard deviations. Flags remain 0/1. The final representation has 17 columns. Constructor identity was a candidate category but was removed after experiments; constructor points and standing remain.

The split is 2003-2018 training (6,411 appearances), 2019-2021 validation (1,200) and 2022-2024 test (1,359). Whole races stay in one split. Form uses only earlier appearances; standings use the previous race within each season. Season-opening points are zero and unknown positions are missing before preprocessing.

Missingness, medians, means and standard deviations are compared before and after preprocessing. All fitting settings come from training; validation supports choices. Figures 19 and 25 show missingness and order comparisons. The setup and selected columns are fixed before examining test scores.

4. Predictive Modeling and Final Test Results

Logistic regression is the simple baseline: additive weighted inputs produce podium log-odds. It can miss nonlinear interactions. Random forest averages 300 trees, with depth 8 and at least 5 samples per leaf; it captures interactions but can still overfit historical patterns. The neural network has ReLU layers of 32 and 16 units, alpha 0.01, batch size 128 and learning rate 0.001; it learns nonlinear combinations but needs scaling and is less interpretable.

The neural checkpoint is selected at epoch 31 using validation PR-AUC, with patience 20 and a maximum of 200 epochs. Final comparisons use the same 17 inputs and appearances for every model. Classification thresholds are selected by validation F1: logistic regression 0.40, random forest 0.48 and neural network 0.43.

PR-AUC is primary because podiums are relatively rare. F1 is secondary because it balances precision and recall. ROC-AUC, precision and recall are also reported. PR-AUC uses trapezoidal integration, not average precision.

Model / split	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic / Training	0.9262	0.6754	0.6494	0.6301	0.6700
Logistic / Validation	0.9295	0.7226	0.7202	0.6748	0.7722
Logistic / Test	0.9345	0.7050	0.6523	0.6385	0.6667
Neural / Training	0.9339	0.6980	0.6569	0.6393	0.6756
Neural / Validation	0.9376	0.7638	0.7335	0.6985	0.7722
Neural / Test	0.9375	0.7078	0.6730	0.6514	0.6961
Forest / Training	0.9558	0.7975	0.6961	0.7391	0.6578
Forest / Validation	0.9391	0.7440	0.7207	0.7247	0.7167
Forest / Test	0.9376	0.7114	0.6564	0.6882	0.6275

The neural network is selected by validation PR-AUC 0.7638, compared with 0.7226 for logistic regression and 0.7440 for random forest. Its test PR-AUC is 0.7078 and test F1 is 0.6730. Random forest has slightly higher test PR-AUC (0.7114); this does not change the preselected model. No choices are retuned using test results.

At threshold 0.43, the network correctly identifies 142 podiums and 1,079 non-podiums, with 62 missed podiums and 76 false podium predictions. Test PR-AUC is lower than validation, showing incomplete transfer to the later period. The predictions are per driver and do not enforce exactly three podium predictions per race. Figures 20-22 show initial comparisons; Figures 27-28 show final results.

5. Experiments

Feature-group ablation retrains the reference neural network after removing one group at a time. The same rows, training duration, seed and fixed threshold are used. Missing-value flags and constructor indicators are removed with their groups. These are individual removals, not a jointly tested combination.

Removed group	Inputs	Validation PR-AUC	PR-AUC drop
None (all features)	46	0.7461	+0.0000
Grid	44	0.7310	+0.0151
Qualifying	44	0.7529	-0.0069
Recent form	41	0.7411	+0.0049
Driver standings	42	0.7434	+0.0027
Constructor standings	42	0.7532	-0.0071
Constructor identity	17	0.7638	-0.0177

Removing grid hurts most (PR-AUC drop 0.0151). Removing constructor identity improves the reference PR-AUC from 0.7461 to 0.7638. We retain all other groups rather than combining individually beneficial removals without testing them. Figure 24 shows the changes.

Two preprocessing orders

Order	Validation PR-AUC	Validation F1
Fill then scale	0.7461	0.7179
Scale then fill	0.7418	0.7194

Fill then scale has higher primary PR-AUC (0.7461 versus 0.7418), although scale then fill has slightly higher F1. We retain fill then scale. Parameters are learned from training in both orders; differences in scale after filling are also documented. Figure 25 presents the comparison.

Final feature confirmation

The full 46-input and reduced 17-input versions are compared for all three models. Reduction does not improve every model, but the reduced network has the highest validation PR-AUC among the tested candidates. All final models use the same reduced representation. Thresholds are refreshed on validation only, then frozen before the test evaluation. Figure 26 shows these comparisons.

The three model families already meet the requirement for at least three distinct modeling attempts. These experiments justify the selected setup; they do not prove that it is optimal for every season or random seed.

6. Model Explainability and Inference

Global explanations

Permutation importance shuffles each input ten times and measures the validation PR-AUC decrease. Qualifying position (0.1527), grid (0.1219) and recent podium rate (0.0857) lead. Kernel SHAP explains 100 validation appearances against 50 training-background examples. Its summary supports the same three leading inputs. Better starting/qualifying positions and higher recent podium rates generally increase predicted podium probability. Figures 29-30 show these results.

One selected appearance

Nikita Mazepin in the 2021 Saudi Arabian Grand Prix has predicted podium probability 0.045%, below the 43% threshold. The model correctly predicts no podium. Grid 20, qualifying 20 and zero recent podiums push SHAP toward a lower prediction. We explain the same appearance with LIME (Figures 31-32).

LIME agrees on the direction of those main factors but assigns other contributions, including to missing-value flags. Its local R^2 is 0.293, a weak local approximation. Its linear surrogate also gives an out-of-range value at this point; it must not be read as the neural network's probability. We report the weak fit rather than treating every LIME weight as reliable.

Inference interface

predict_podium(record) accepts the nine original-unit numeric features. It rejects unexpected fields, treats grid zero as unavailable, creates flags before filling, reuses training medians and scaler, orders the 17 columns, and returns probability, class, threshold and filled fields. It does not retrain or read the current outcome.

Demonstration	Podium probability	Prediction	Filled fields
Complete record	68.60%	Podium	None
Partially missing record	13.33%	No podium	qualifying_position, recent_avg_finish

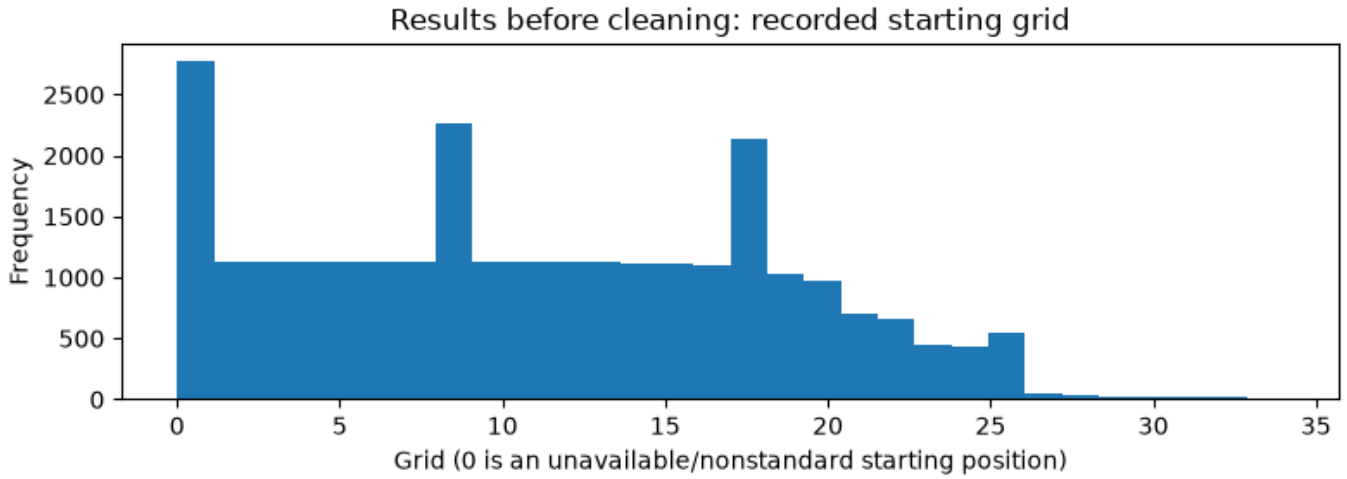
The complete record matches the existing modeling pipeline. The partially missing example withholds qualifying position and recent average finish, so it receives the existing training medians and flags. A valid prediction does not recover the true unknown values, and this two-record demonstration is not a performance estimate.

Limitations and reproducibility

These are model explanations, not causal effects. Related inputs can share information. Historical missingness, season changes, conservative failure labels and unresolved cleaning flags limit interpretation. All 193 code cells passed a fresh local execution; the original notebook and raw CSV hashes were unchanged. The notebook, dependency record, figures and validation evidence accompany this report.

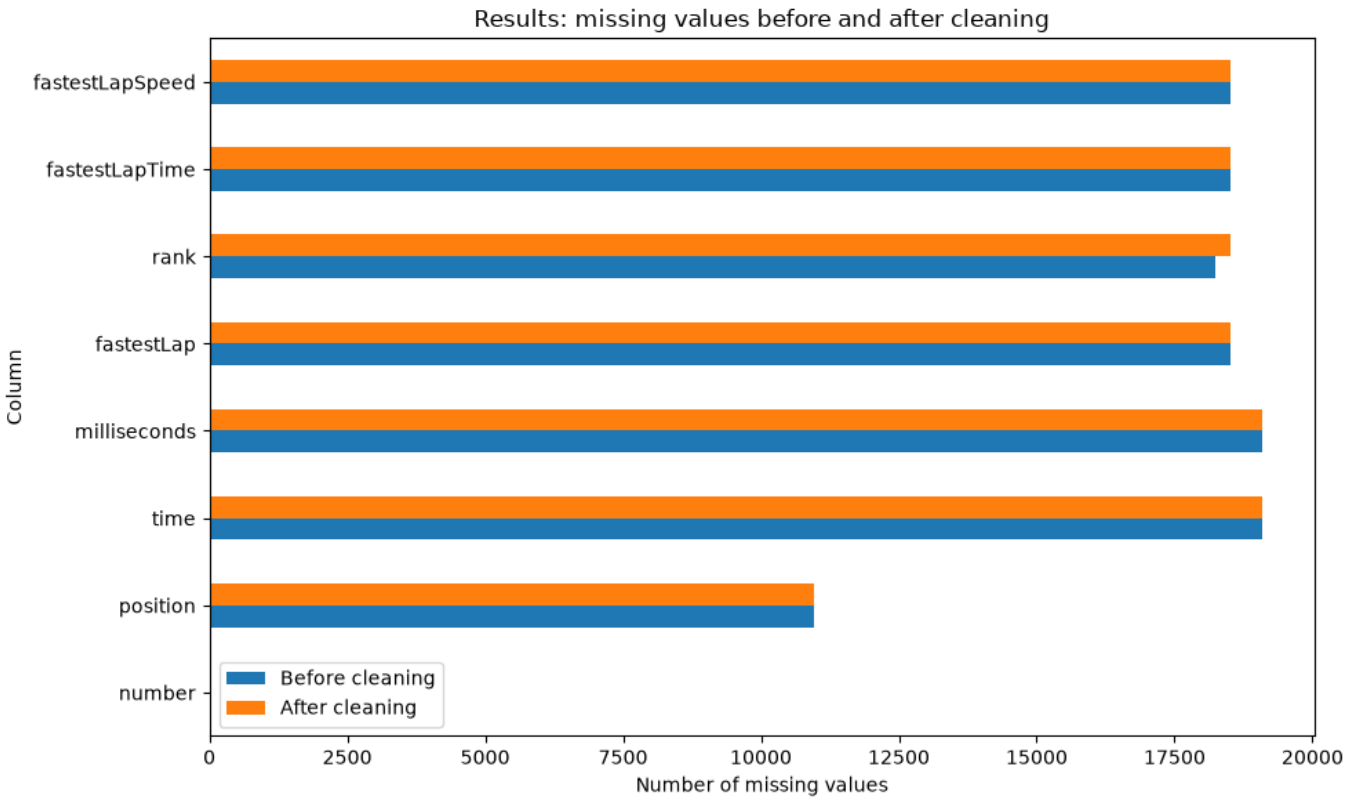
Figure Appendix

Figure 1. Recorded starting grids before cleaning



Source: executed Notebook0, cell 9.

Figure 2. Results missing values before and after cleaning



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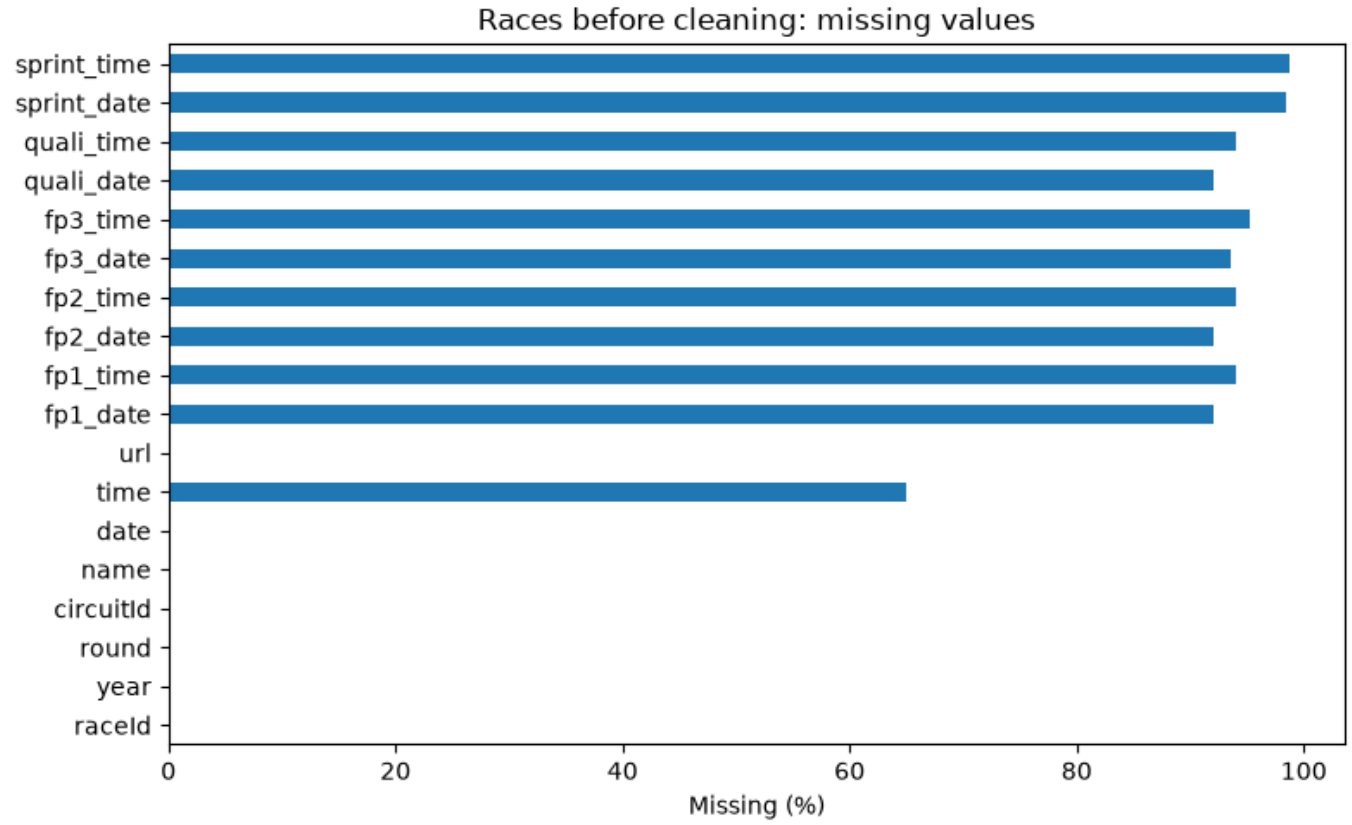
Figure Appendix

Figure 3. Driver missingness before cleaning



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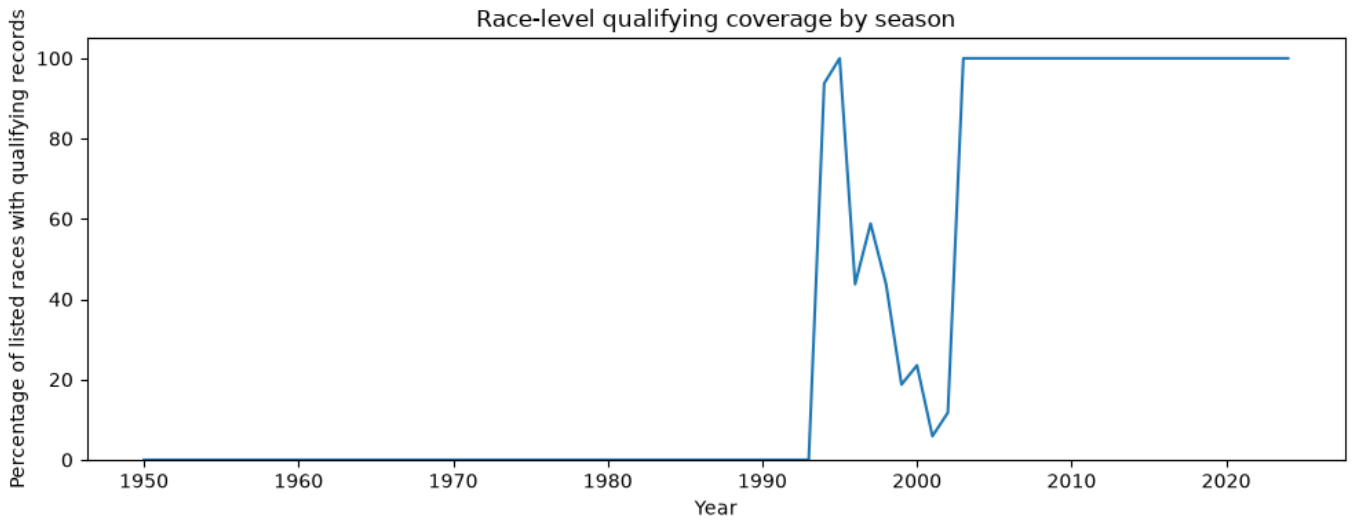
Figure 4. Race missingness before cleaning



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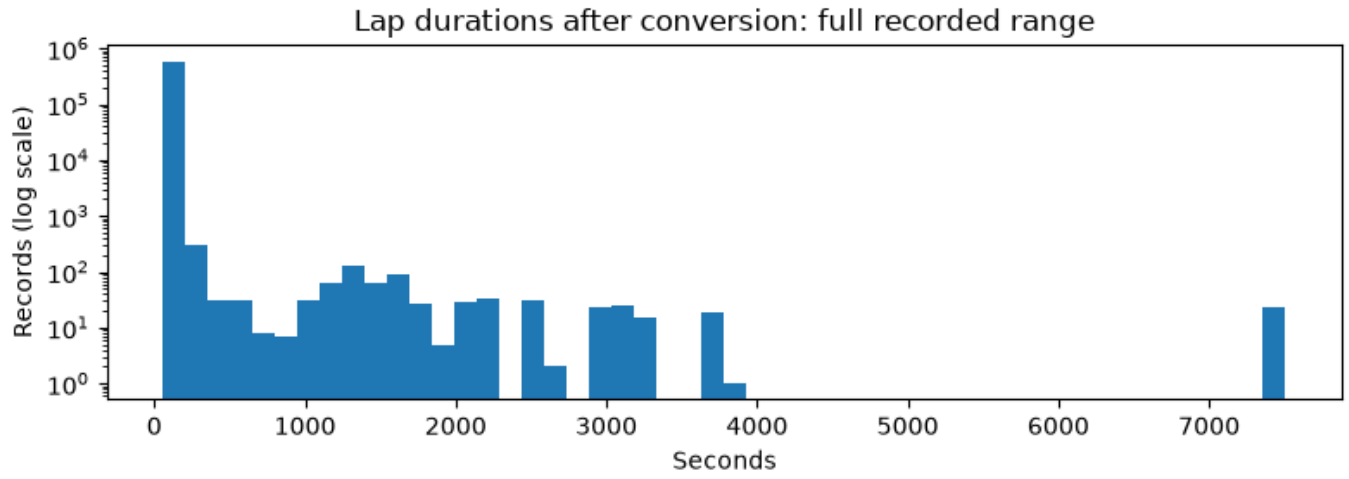
Figure Appendix

Figure 5. Race-level qualifying coverage by season



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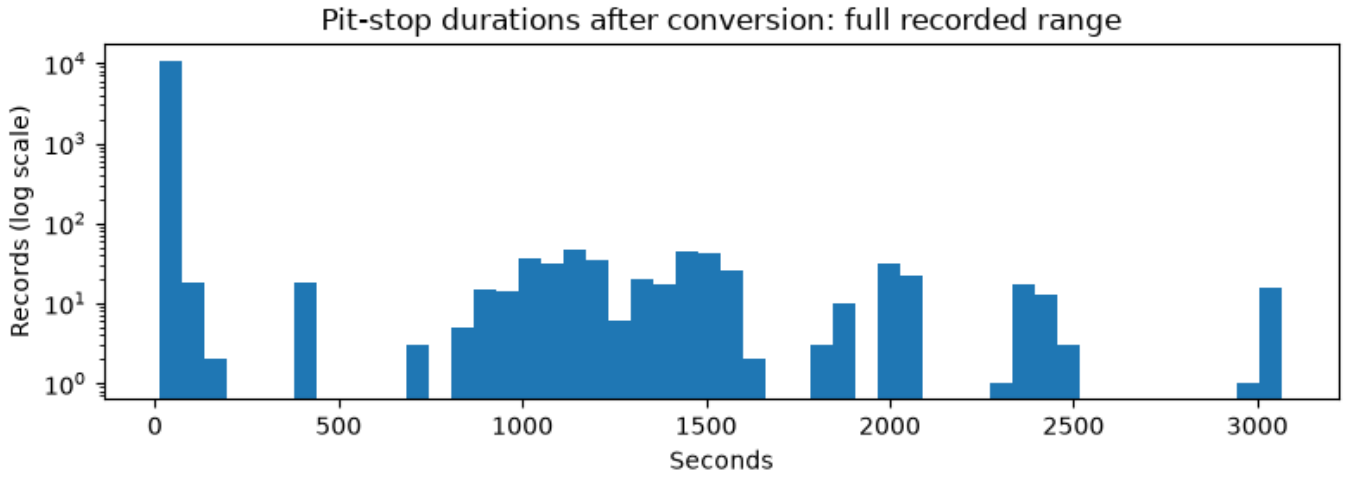
Figure 6. Full recorded lap-duration range



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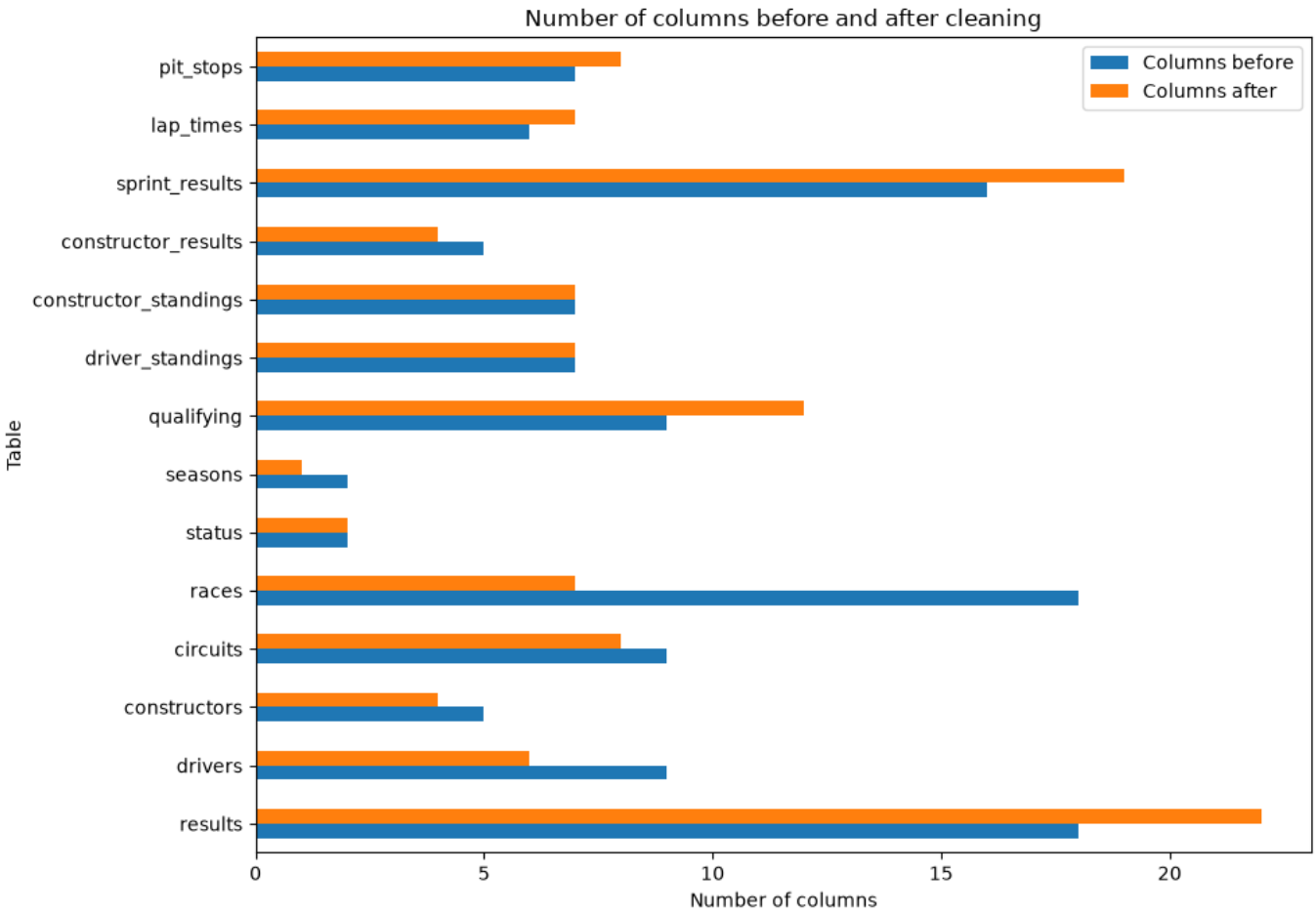
Figure Appendix

Figure 7. Full recorded pit-stop-duration range



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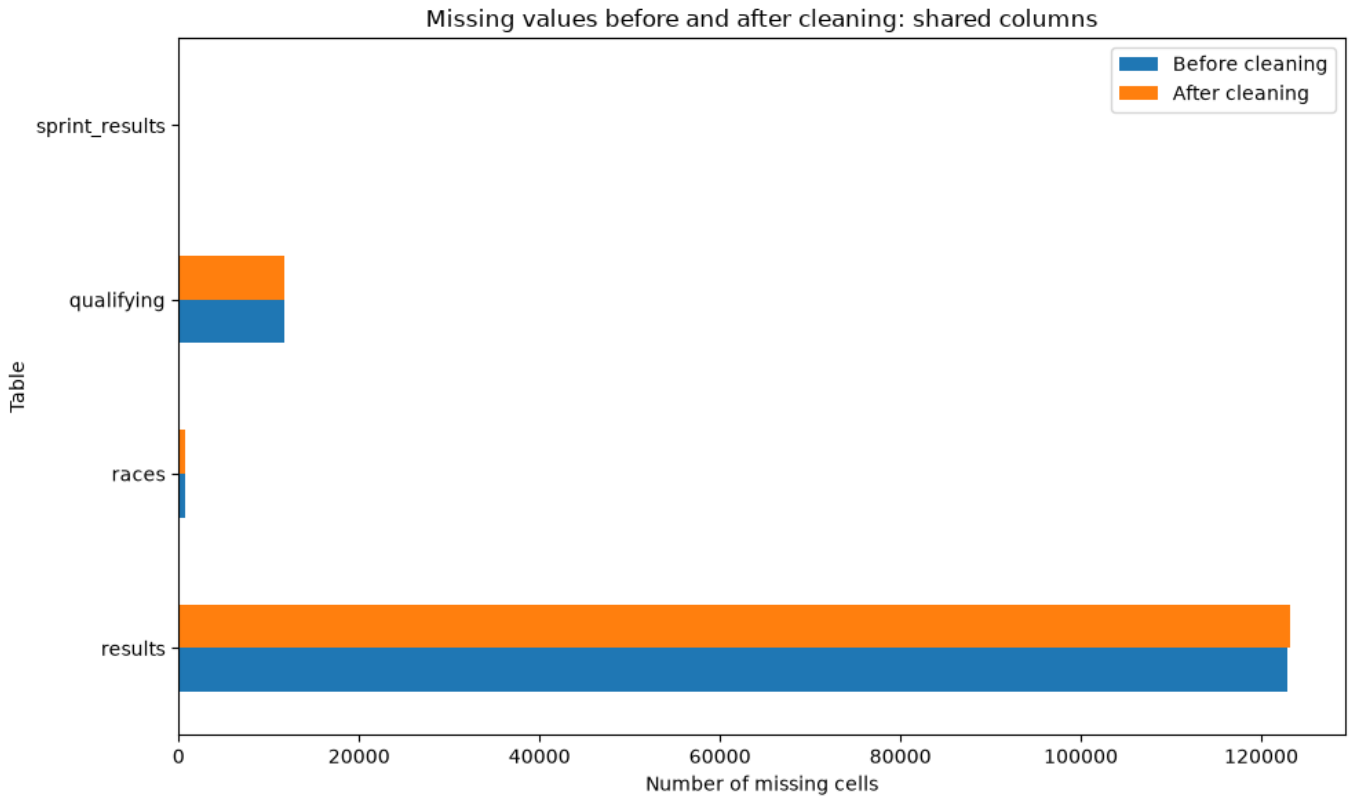
Figure 8. Column counts before and after cleaning



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Figure Appendix

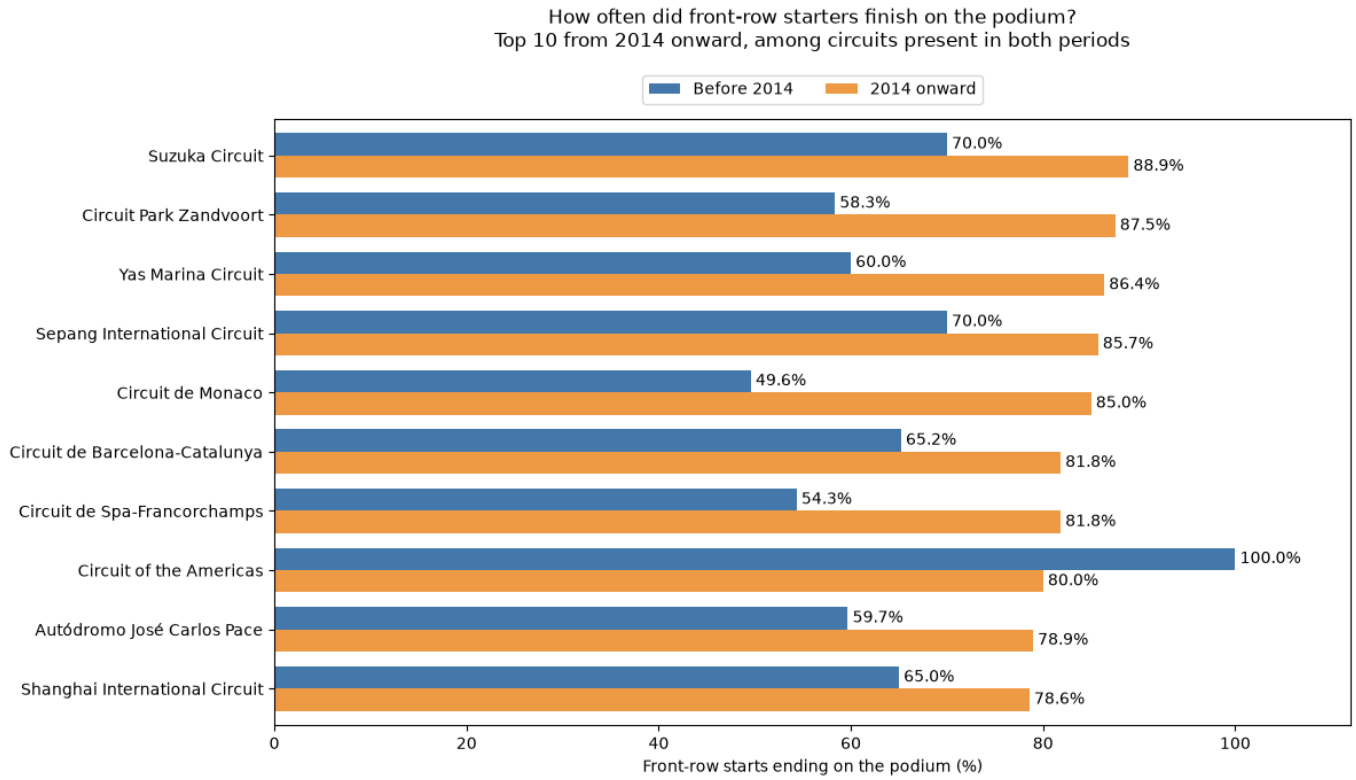
Figure 9. Missing cells before and after cleaning in shared columns



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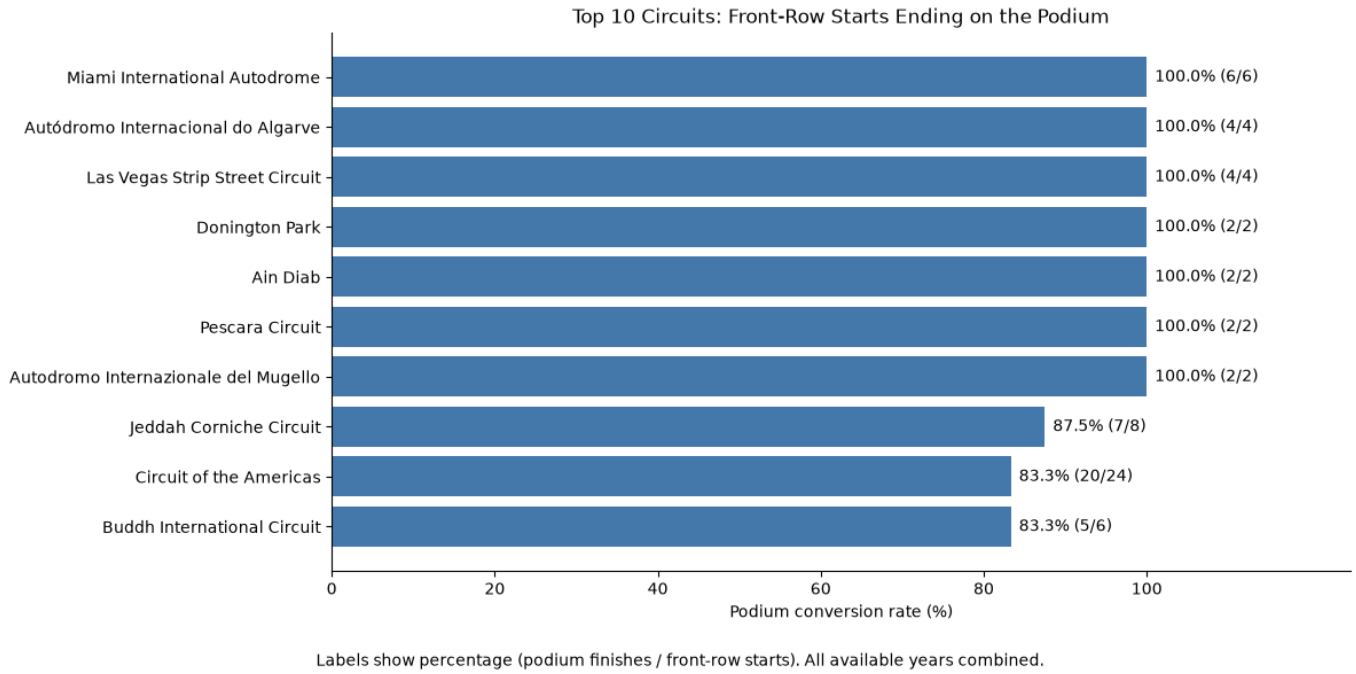
Figure Appendix

Figure 10. Question 1: circuit conversion before and after 2014



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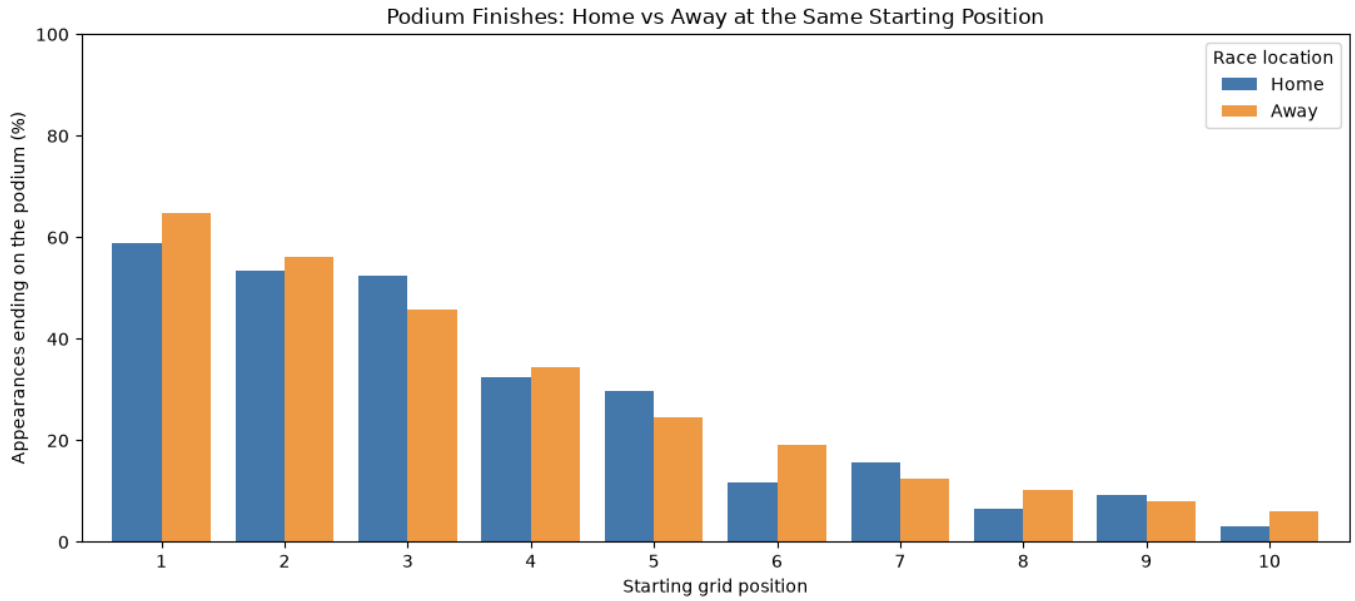
Figure 11. Question 1: highest overall conversion rates and sample sizes



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Figure Appendix

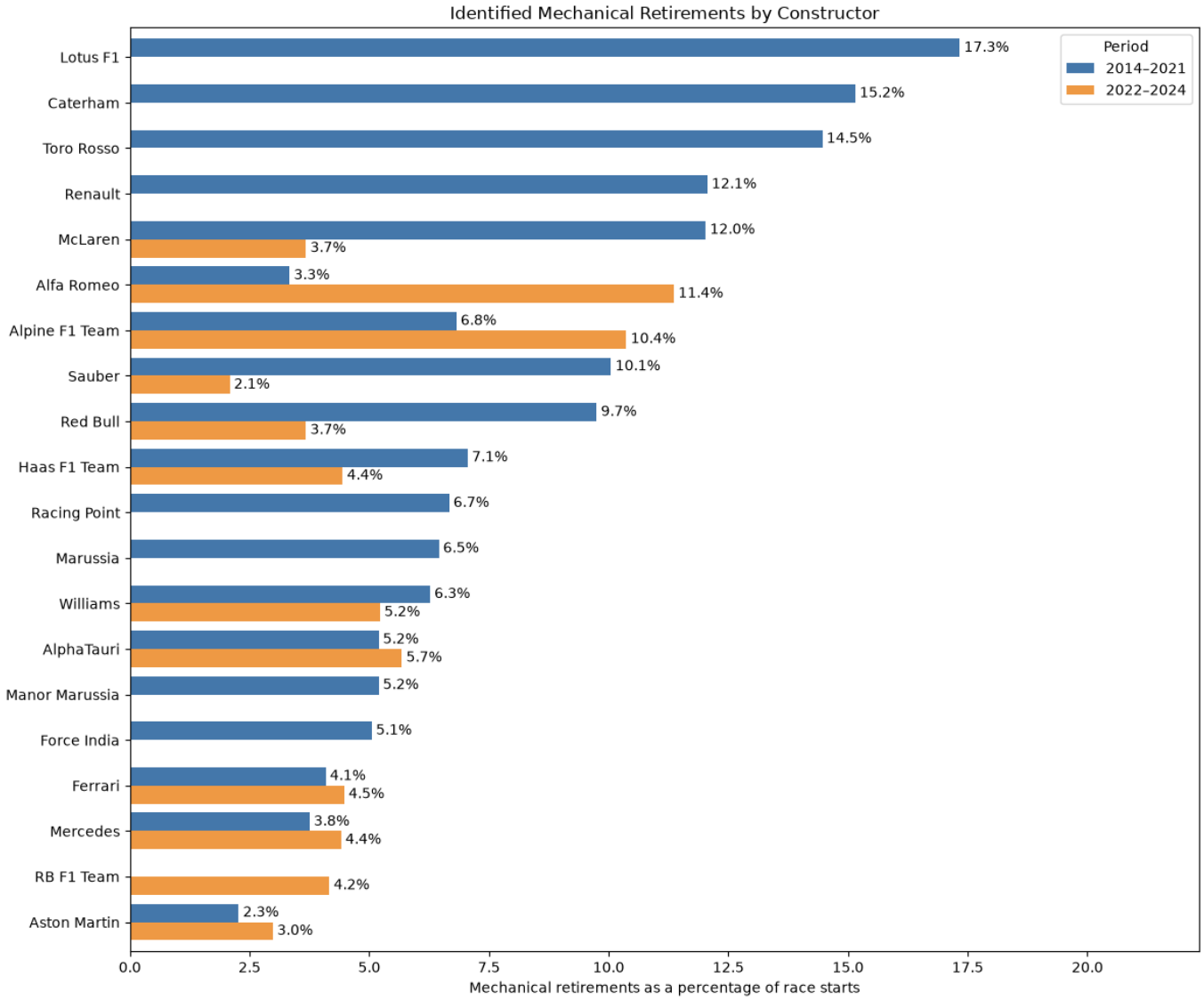
Figure 12. Question 2: home versus away at the same starting grid



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Figure Appendix

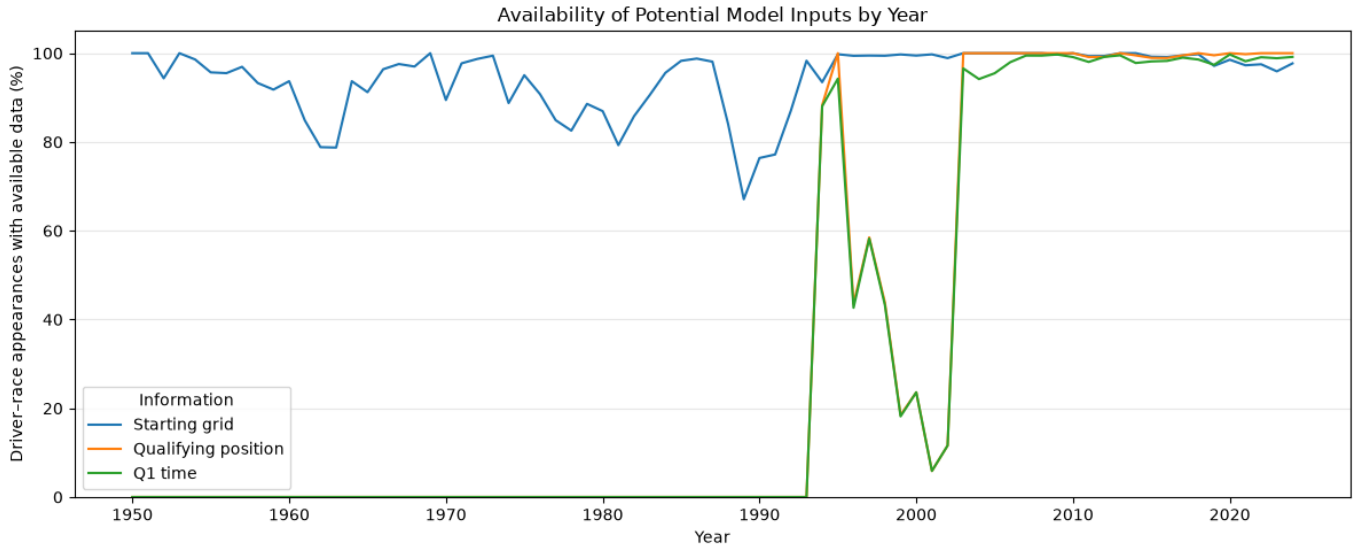
Figure 13. Question 3: identified mechanical retirements by constructor



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Figure Appendix

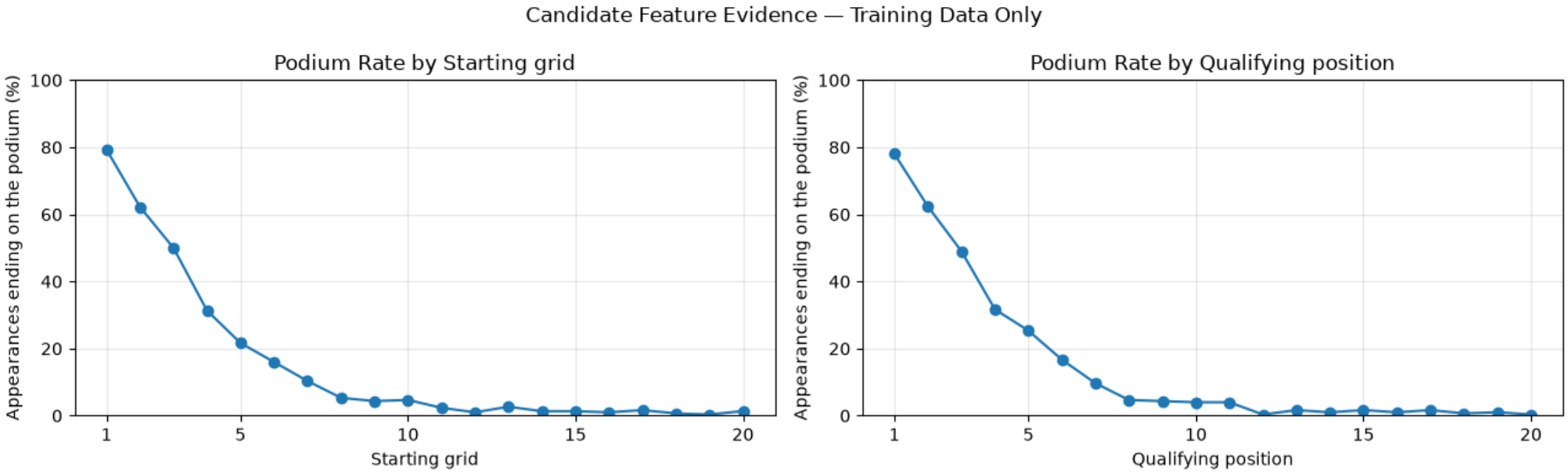
Figure 14. Availability of candidate inputs by year



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Figure Appendix

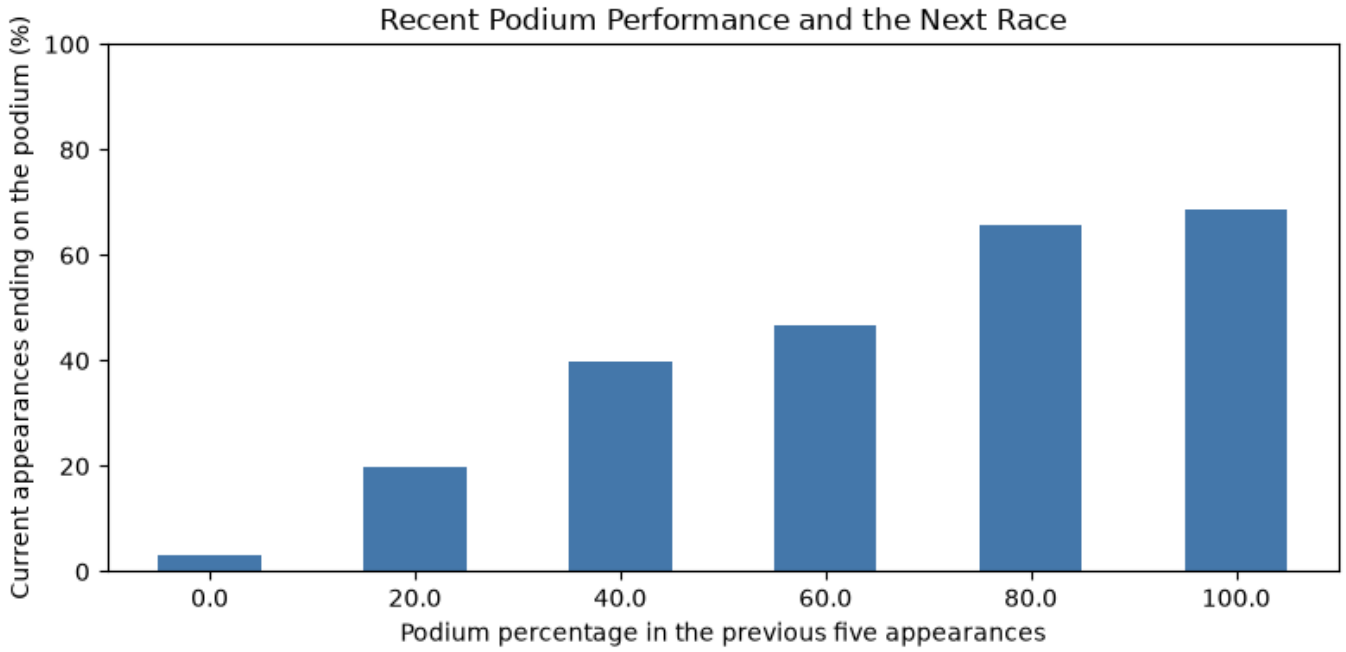
Figure 15. Training evidence: starting grid and qualifying position



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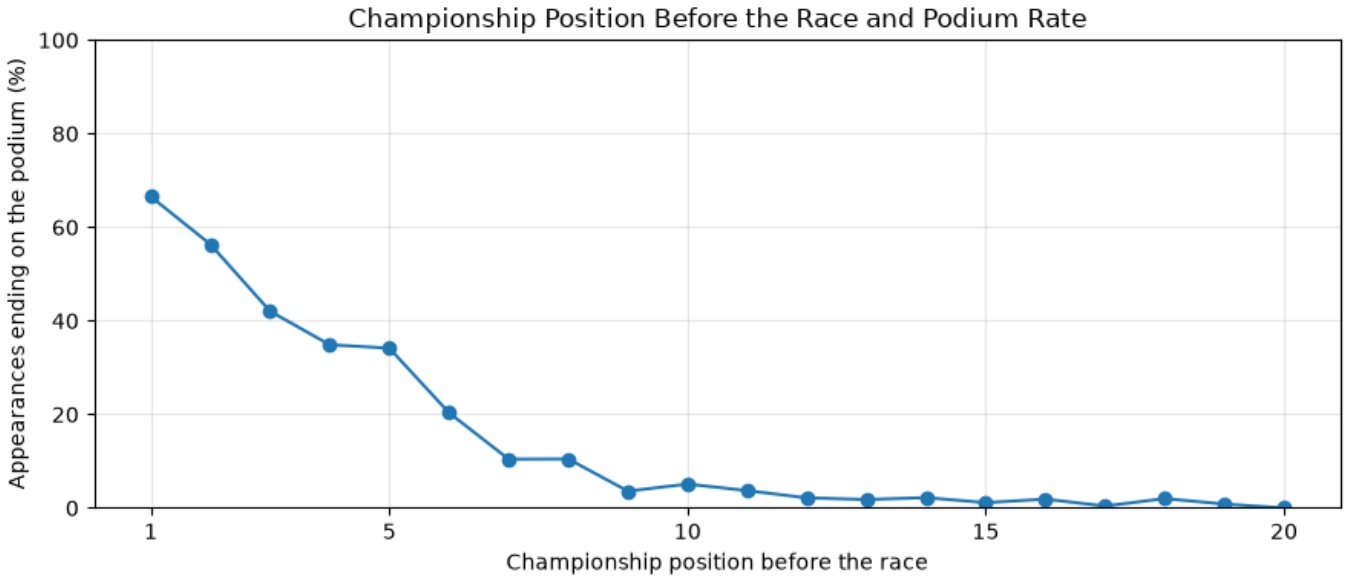
Figure Appendix

Figure 16. Training evidence: recent podium performance



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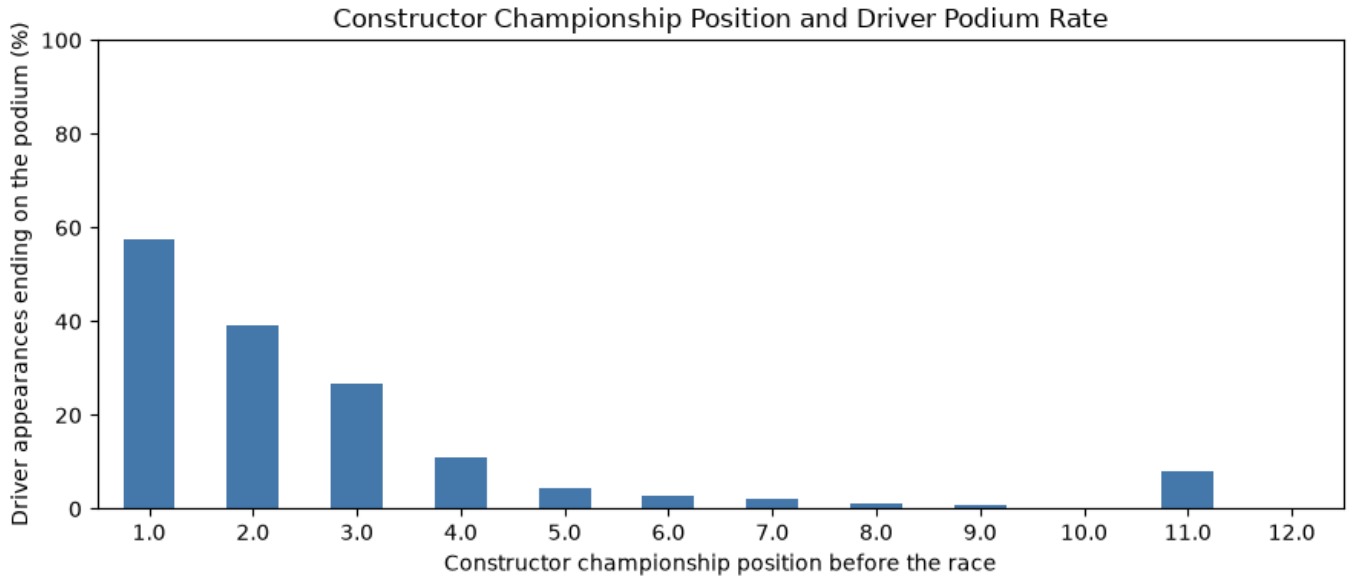
Figure 17. Training evidence: driver standing before the race



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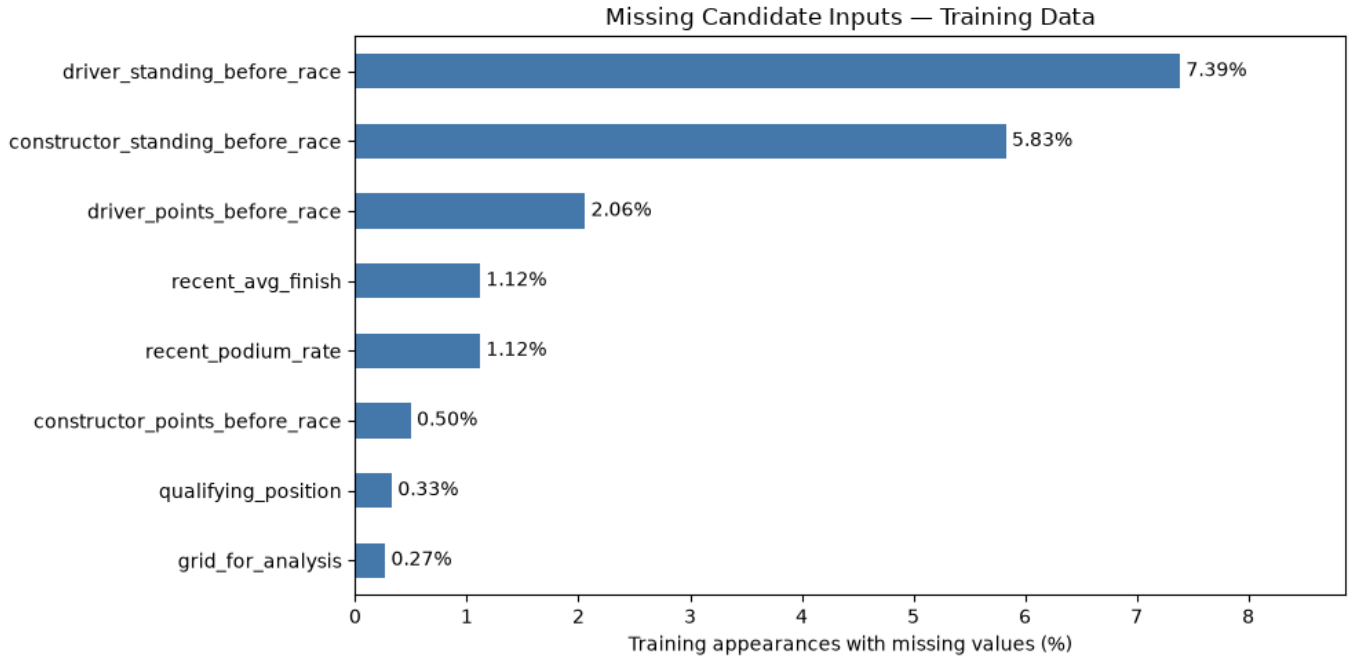
Figure Appendix

Figure 18. Training evidence: constructor standing before the race



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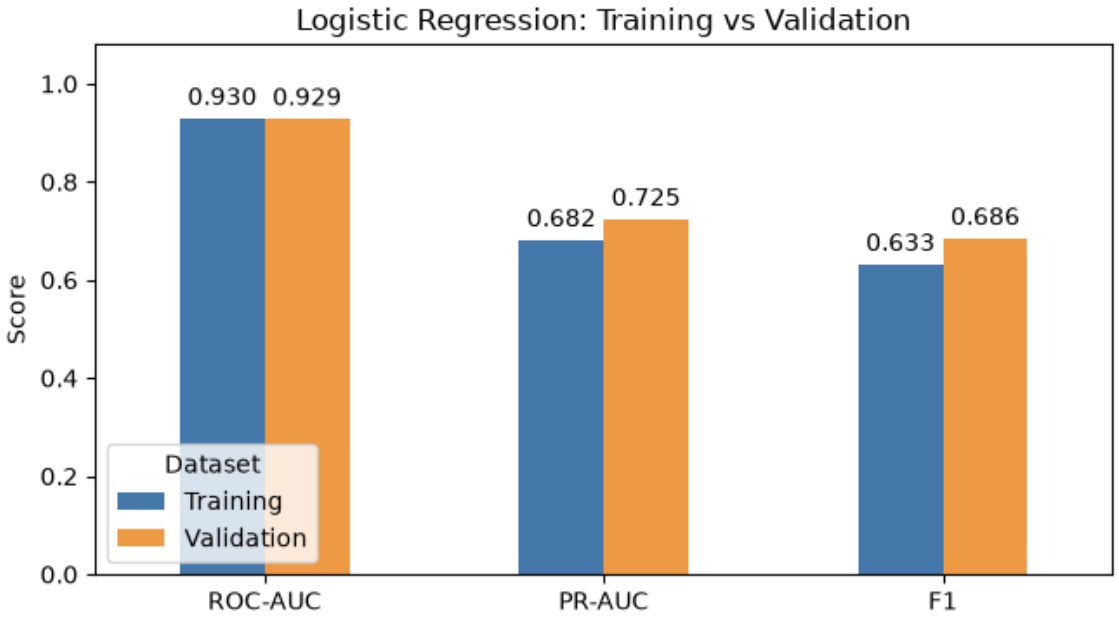
Figure 19. Training missingness of candidate inputs

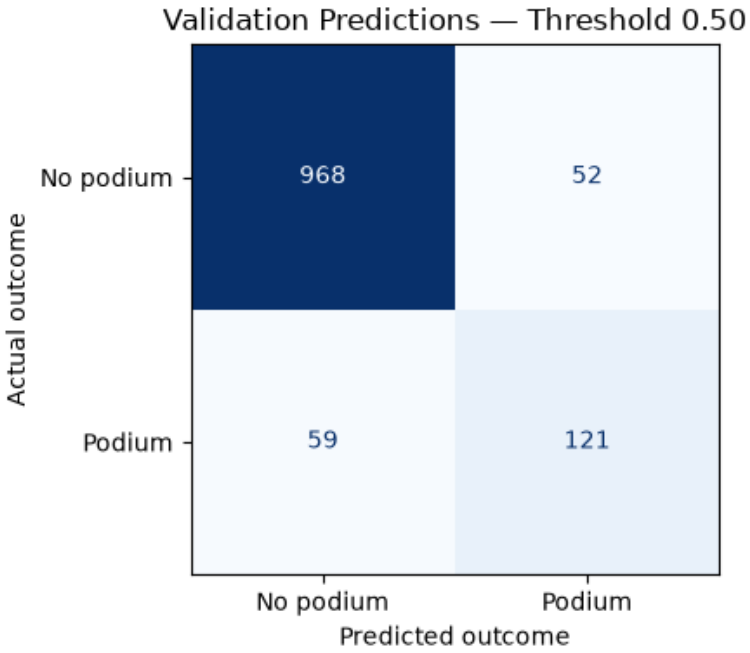


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Figure Appendix

Figure 20. Initial logistic regression: training, validation and confusion matrix

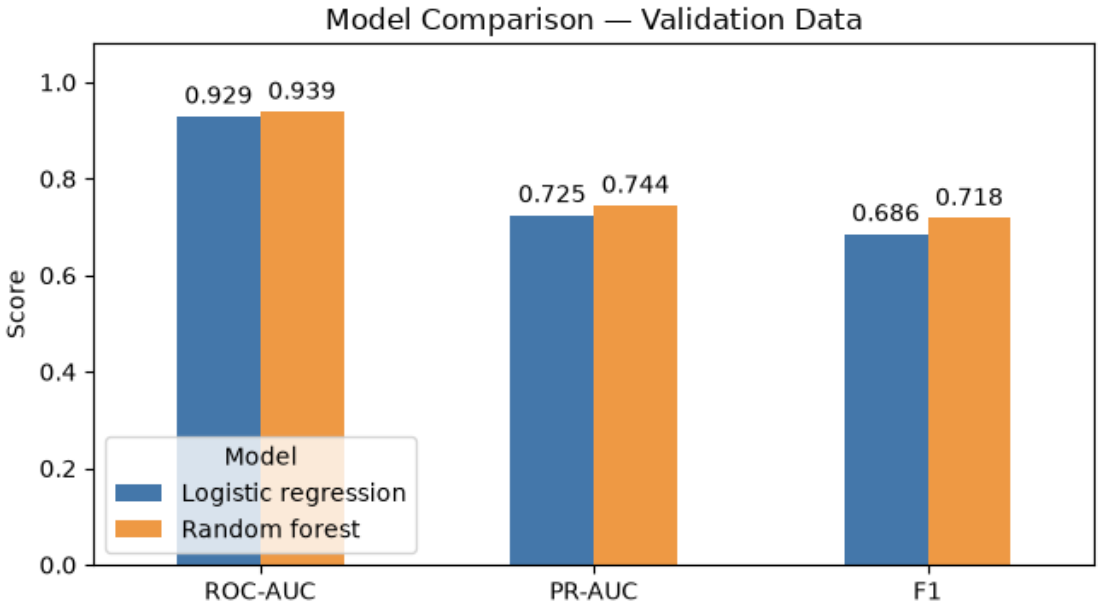


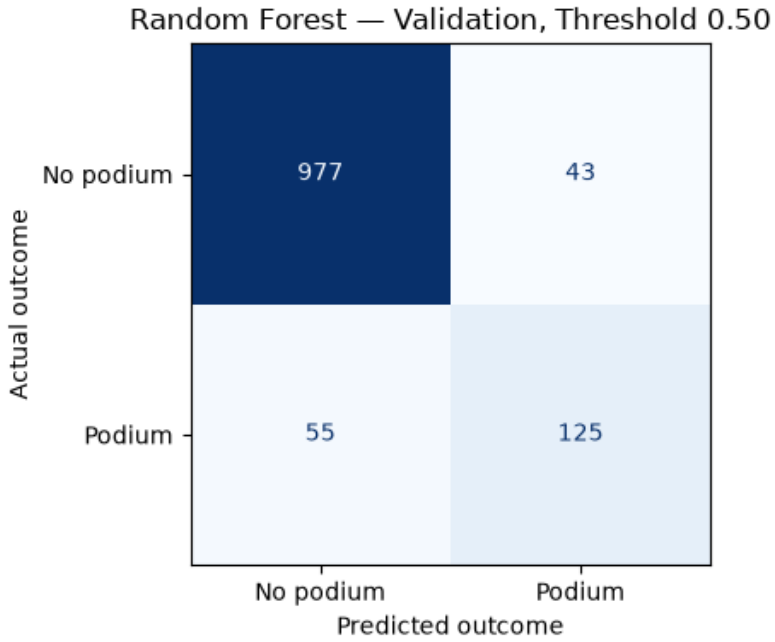


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Figure Appendix

Figure 21. Initial random forest comparison and confusion matrix

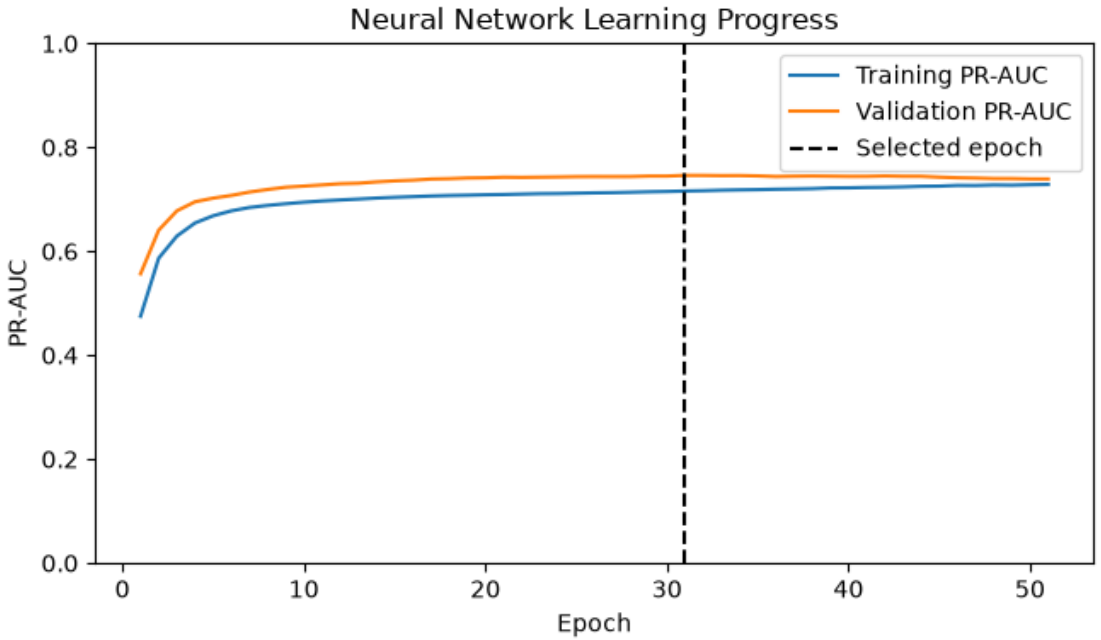


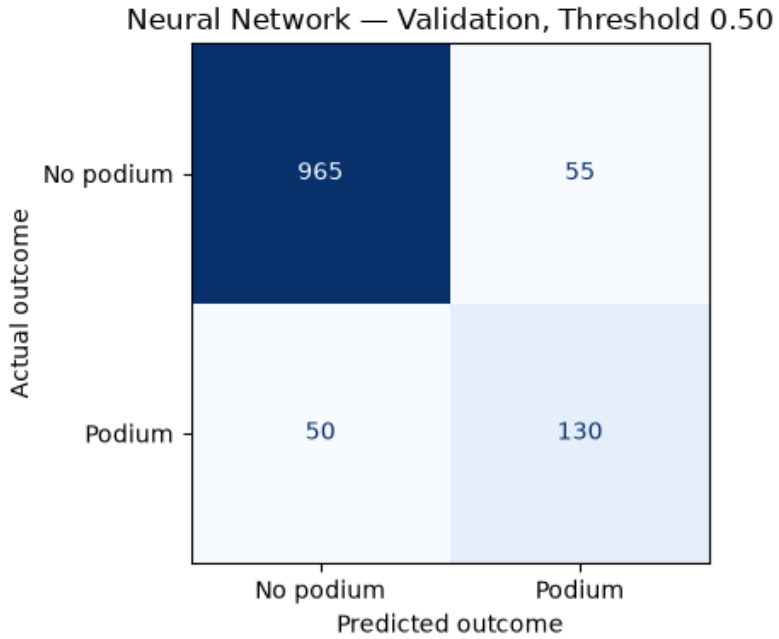


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Figure Appendix

Figure 22. Initial neural network learning curve and confusion matrix

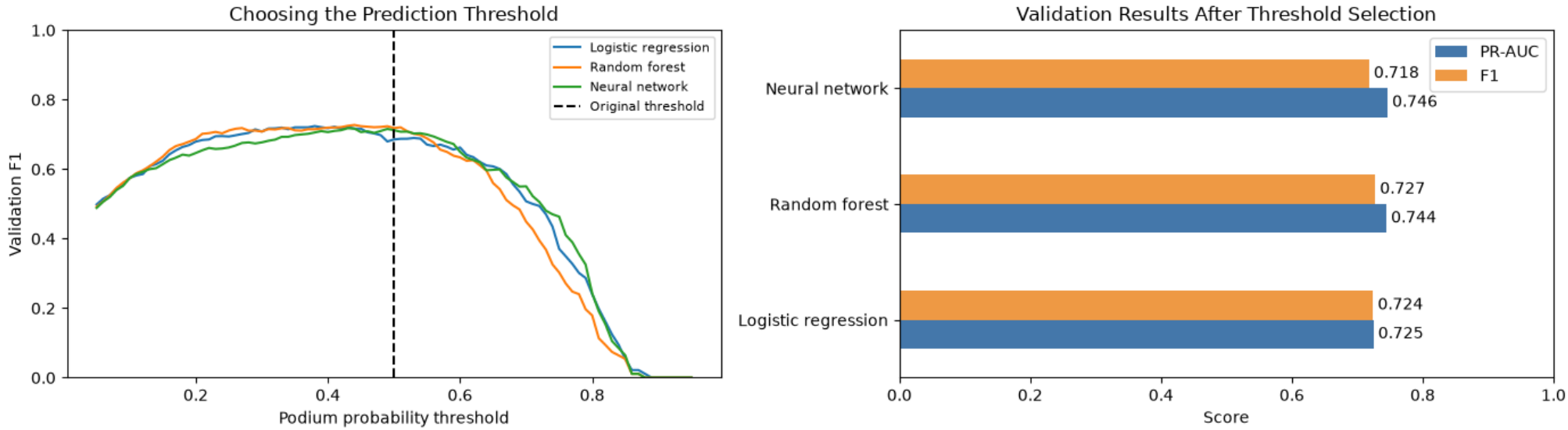




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Figure Appendix

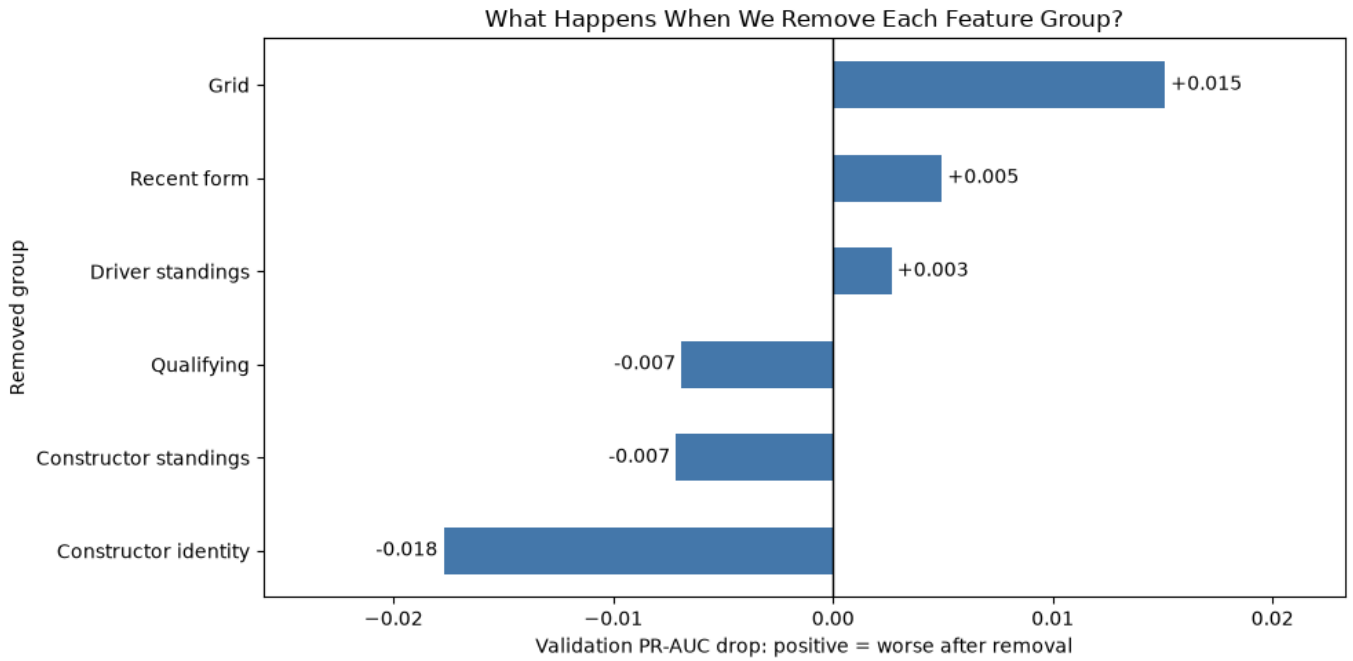
Figure 23. Validation threshold selection and initial model comparison



Source: executed Notebook0, cell 224. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

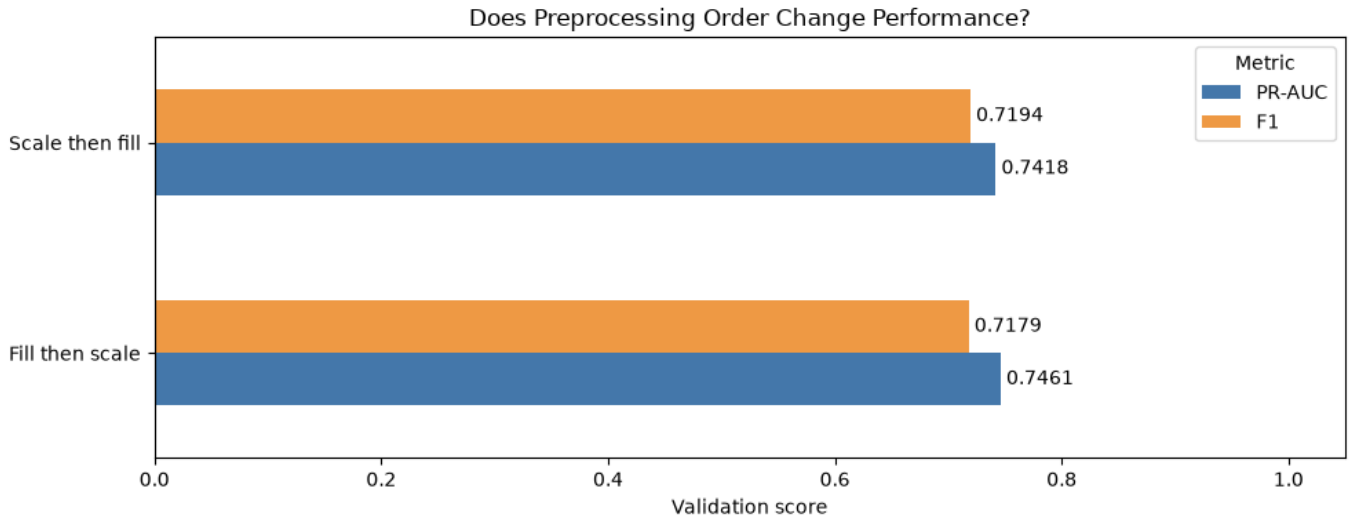
Figure Appendix

Figure 24. Feature-group ablation



Source: executed Notebook0, cell 231. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

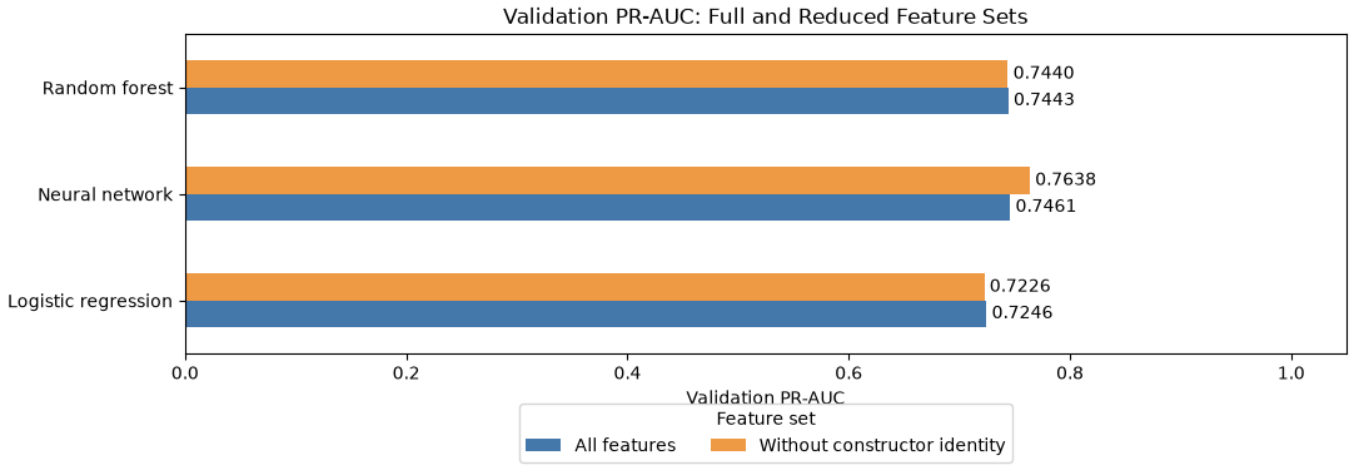
Figure 25. Preprocessing-order experiment



Source: executed Notebook0, cell 234. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

Figure Appendix

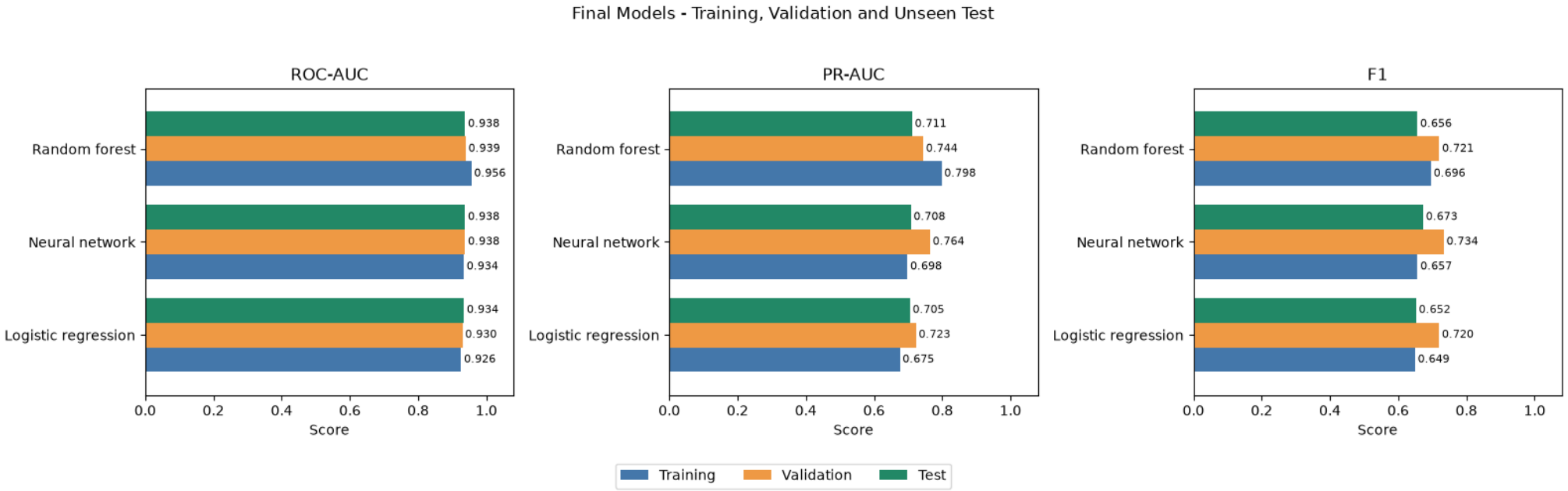
Figure 26. Full versus reduced feature confirmation



Source: executed Notebook0, cell 238. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

Figure Appendix

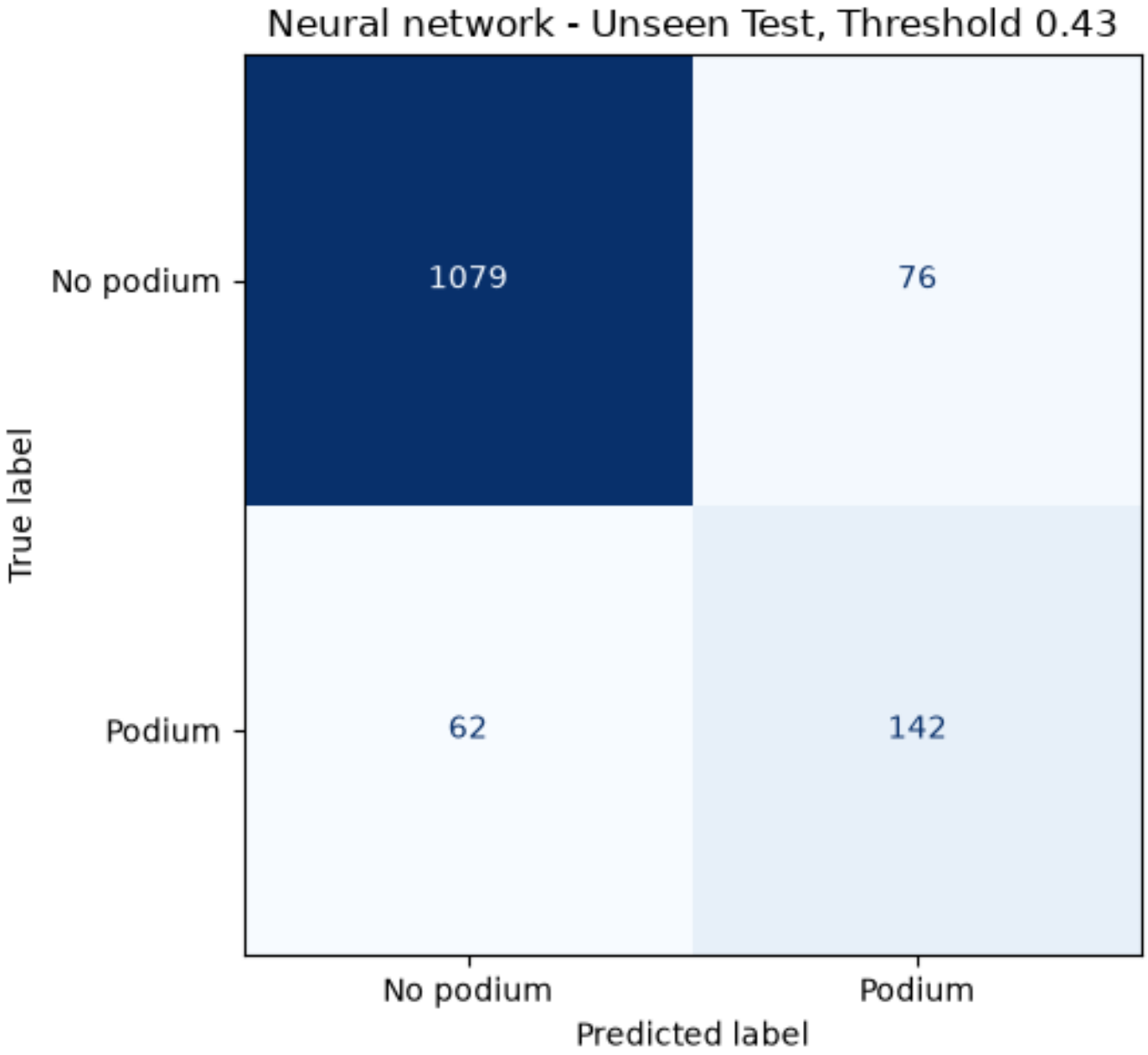
Figure 27. Final model comparison on training, validation and unseen test



Source: executed Notebook0, cell 245. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

Figure Appendix

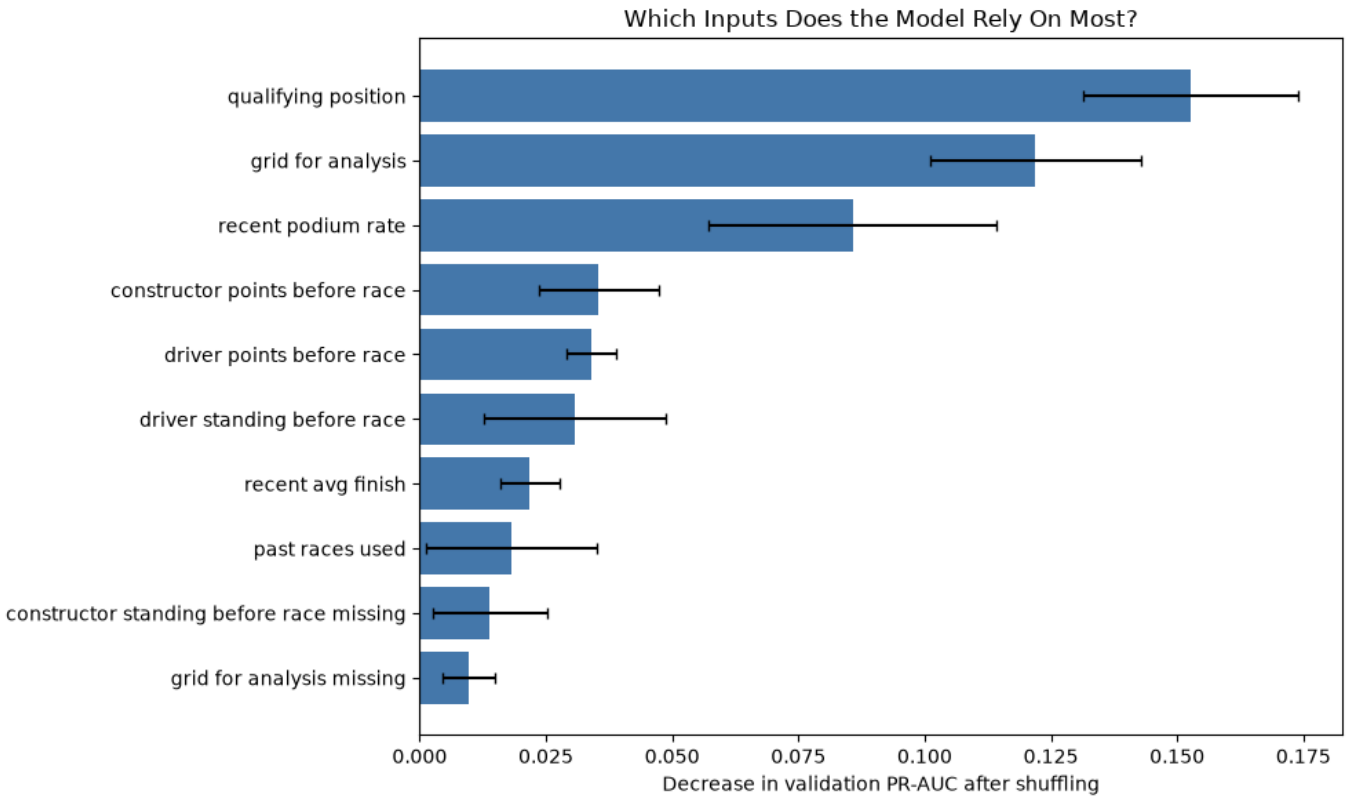
Figure 28. Selected neural network test confusion matrix



Source: executed Notebook0, cell 245. Initial 46-input results are explicitly labeled; final model results use 17 inputs.

Figure Appendix

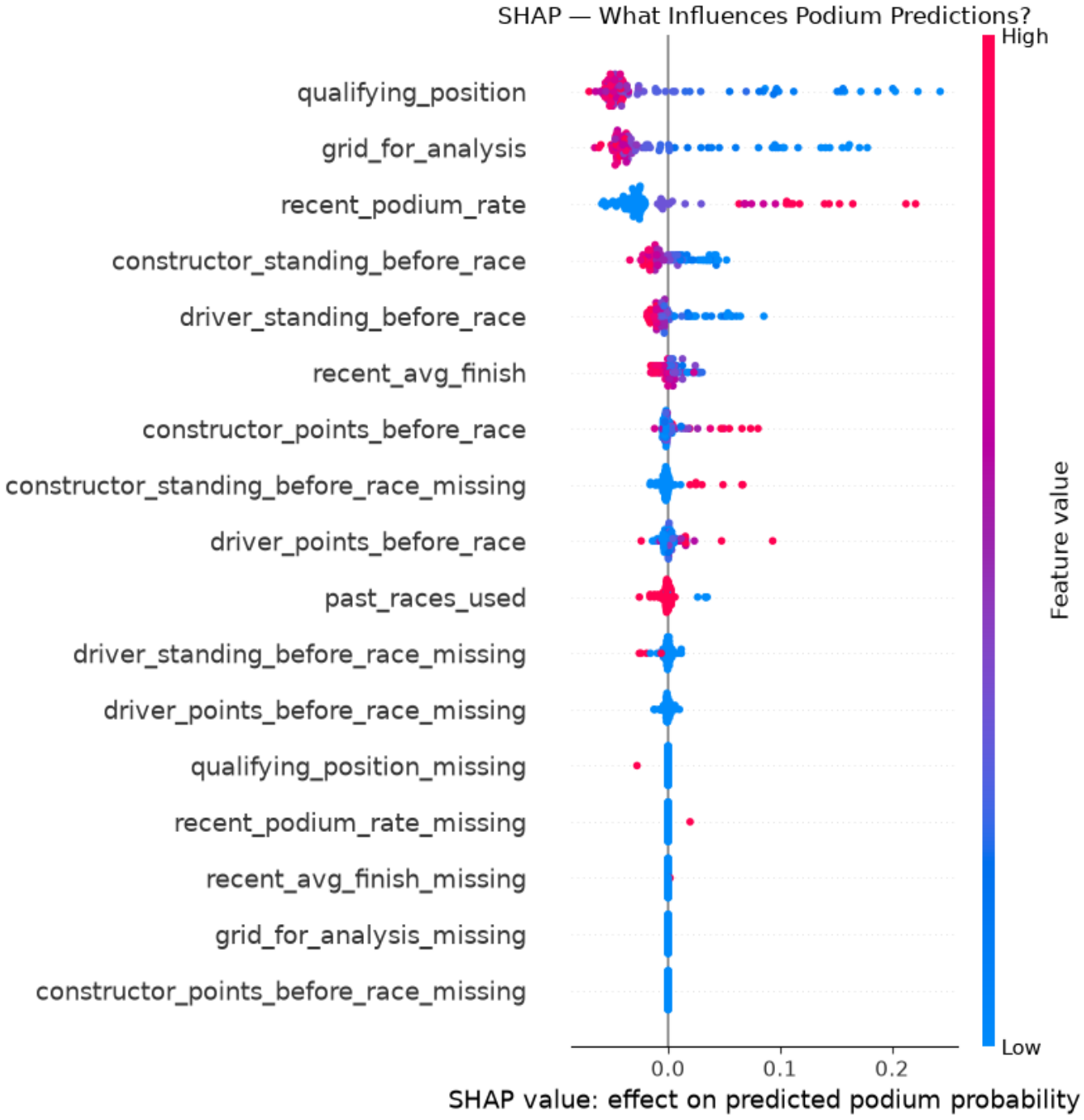
Figure 29. Global permutation importance



Source: executed Notebook0, cell 255.

Figure Appendix

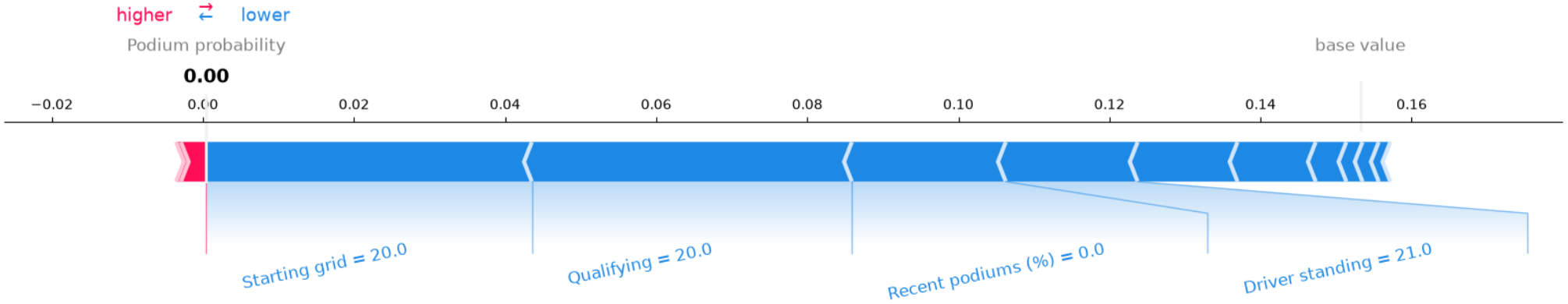
Figure 30. Global SHAP summary



Source: executed Notebook0, cell 256.

Figure Appendix

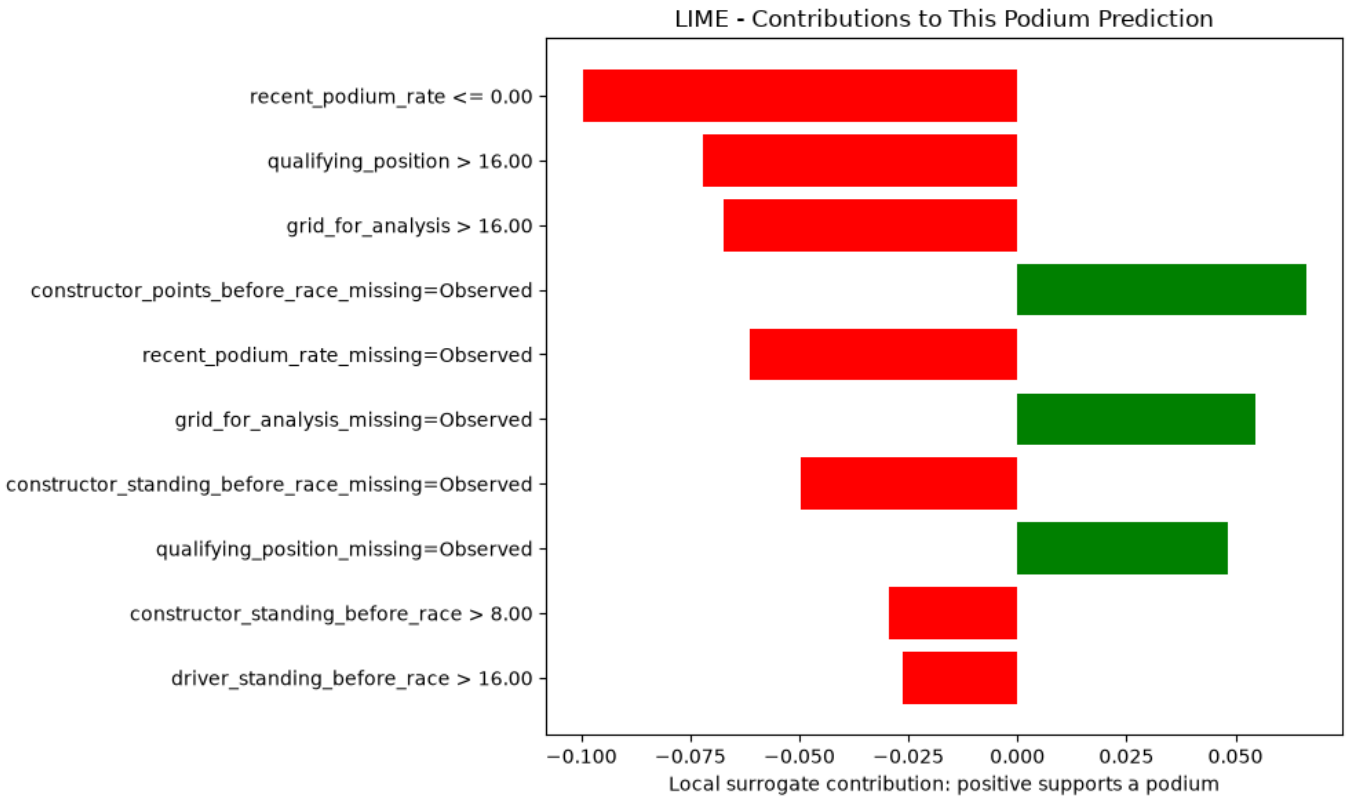
Figure 31. Local SHAP force plot



Source: executed Notebook0, cell 257.

Figure Appendix

Figure 32. Local LIME explanation



Source: executed Notebook0, cell 258.